$$n_1 \frac{DB}{DO} = n_2 \frac{DB}{DI}$$

$DB$ is cancelled both sides. Now, $DO$ is the actual depth $d$ and $DI$ is the apparent depth $d'$.

$$n_1 \frac{1}{d} = n_2 \frac{1}{d'}$$ After rearranging,

$$\frac{d'}{d} = \frac{n_2}{n_1} \qquad (6.25)$$

Rewriting the above equation for the apparent depth $d'$,

$$d' = \frac{n_2}{n_1} d \qquad (6.26)$$

As the rarer medium is air, its refractive index $n_2$ can be taken as 1, ($n_2$=1). And the refractive index $n_1$ of denser medium could then be taken as $n$ itself, ($n_1$=$n$). Now, the equation for apparent depth becomes,

$$d' = \frac{d}{n} \qquad (6.27)$$

The bottom appears to be elevated by $d$-$d'$,

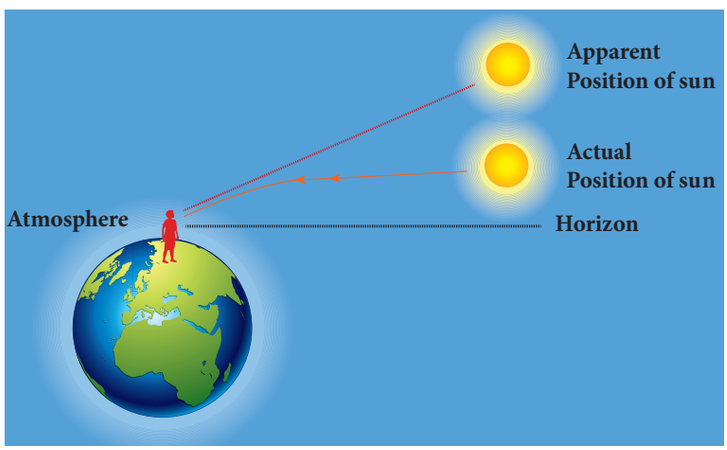
$$d - d' = d - \frac{d}{n} \text{ or}$$

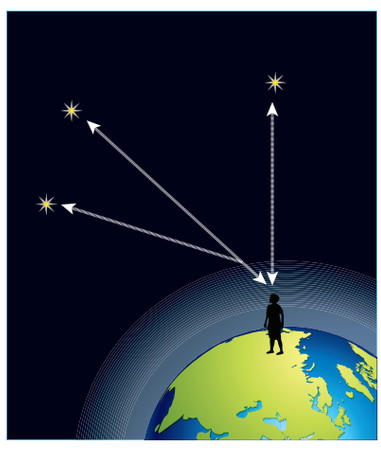
$$d - d' = d\left(1 - \frac{1}{n}\right) \qquad (6.28)$$




**Atmospheric refraction:** Due to refraction of light through different layers of atmosphere which vary in refractive index, the path of light deviates continuously when it passes through the atmosphere. For example, the Sun is visible a little before the actual sunrise and also until a little after the actual sunset due to refraction of light through the atmosphere. What we mean by actual sunrise is the actual crossing of the sun at the horizon. Figure shows the actual and apparent positions of the sun with respect to the horizon. The figure is highly exaggerated to show the effect. The apparent shift in the direction of the sun is around half a degree and the corresponding time difference between the actual and apparent positions is about 2 minutes. Sun appears flattened (oval shaped) during sun rise and sunset due to the same phenomenon.

The same is also applicable for the positions of stars as shown in Figure. Actually, the stars do not twinkle. They appear twinkling because of the movement of the atmospheric layers with varying refractive indices which is clearly seen in the night sky.





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